

Laporan Praktikum
Pemrograman Web 2
Pertemuan 14
Deployment, Docker & CI/CD



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Bab 1. Latihan

1. Membuat Container untuk website job portal menggunakan Docker.

Pertama, kita ubah konfigurasi database di .env agar bisa sesuai dengan konfigurasi docker.

```
1 DB_CONNECTION=mysql
2 DB_HOST=db
3 DB_PORT=3306
4 DB_DATABASE=jobportal
5 DB_USERNAME=root
6 DB_PASSWORD=root
```

Selanjutnya, dibuat **Dockerfile** yang berisi:

```
1 FROM php:8.2-fpm
2
3 # Install system dependencies
4 RUN apt-get update && apt-get install -y \
5     libpng-dev \
6     libjpeg-dev \
7     libfreetype6-dev \
8     libzip-dev \
9     zip \
10    unzip \
11    git \
12    curl \
13    && docker-php-ext-configure gd --with-freetype --
14    with-jpeg \
15    && docker-php-ext-install pdo pdo_mysql gd zip \
16    && rm -rf /var/lib/apt/lists/*
17
18 ENV COMPOSER_ALLOW_SUPERUSER=1
19
20 # Set working directory
21 WORKDIR /var/www
22
23 # Copy project files
24 COPY . .
25
26 # Install Composer dependencies
27 COPY --from=composer:2.6 /usr/bin/composer /usr/bin/
28 composer
29 RUN composer install --no-dev --optimize-autoloader
```

```
28
29 # Create necessary directories and set permissions
30 RUN mkdir -p /var/www/storage /var/www/bootstrap/cache
31 \
32   && chown -R www-data:www-data /var/www/storage /
33   var/www/bootstrap/cache
34 EXPOSE 9000
35 CMD ["php-fpm"]
```

Lalu dalam docker-compose.yml:

```
1 services:
2   app:
3     build:
4       context: .
5       dockerfile: Dockerfile
6     container_name: laravel-app
7     restart: always
8     working_dir: /var/www
9     volumes:
10      - ./var/www
11     depends_on:
12      - db
13     environment:
14      - DB_CONNECTION=mysql
15      - DB_HOST=db
16      - DB_PORT=3306
17      - DB_DATABASE=jobportal
18      - DB_USERNAME=root
19      - DB_PASSWORD=root
20
21   webserver:
22     image: nginx:alpine
23     container_name: nginx-server
24     restart: always
25     ports:
26      - "8000:80"
27     volumes:
28      - ./var/www
29      - ./nginx/conf.d:/etc/nginx/conf.d
30     depends_on:
31      - app
32
33   db:
34     image: mysql:8.0
35     container_name: mysql-db
36     restart: always
37     environment:
38      MYSQL_DATABASE: jobportal
39      MYSQL_ROOT_PASSWORD: root
```

```
40     MYSQL_ROOT_HOST: "%"
41     ports:
42     - "3307:3306"
43     volumes:
44     - db_data:/var/lib/mysql
45
46     phpmyadmin:
47     image: phpmyadmin/phpmyadmin
48     container_name: phpmyadmin
49     restart: always
50     ports:
51     - "8080:80"
52     environment:
53     PMA_HOST: db
54     PMA_USER: root
55     PMA_PASSWORD: root
56     depends_on:
57     - db
58
59     volumes:
60     db_data:
```

Karena menggunakan MySQL, kita juga bisa menambahkan UI untuk phpmyadmin jika diinginkan dengan dengan cara diatas.

Terakhir untuk config nginx:

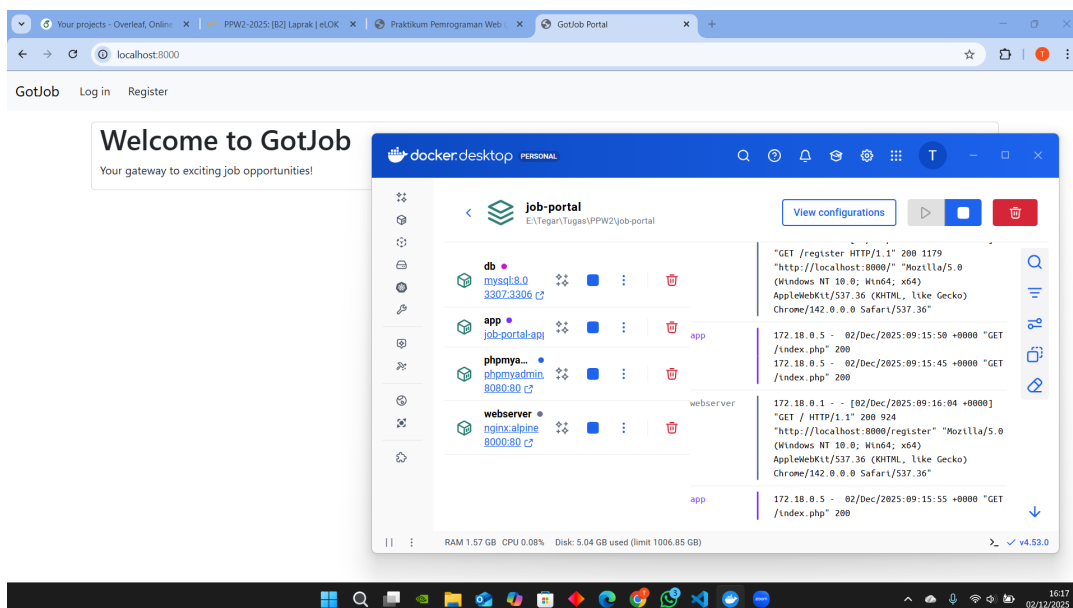
```
1  server {
2      listen 80;
3      server_name localhost;
4      root /var/www/public;
5      index index.php index.html;
6
7      # Logs
8      access_log /var/log/nginx/access.log;
9      error_log /var/log/nginx/error.log;
10
11     # Security headers
12     add_header X-Frame-Options "SAMEORIGIN" always;
13     add_header X-Content-Type-Options "nosniff" always
14     ;
15     add_header X-XSS-Protection "1; mode=block" always
16     ;
17
18     # Gzip compression
19     gzip on;
20     gzip_types text/plain text/css text/javascript
21     application/javascript;
22
23     # Laravel routing
24     location / {
```

```

22         try_files $uri $uri/ /index.php?$query_string;
23     }
24
25     # Pass PHP requests to the PHP-FPM container
26     location ~ /\.php$ {
27         try_files $uri =404;
28         fastcgi_pass app:9000;
29         fastcgi_index index.php;
30         fastcgi_param SCRIPT_FILENAME
31             $document_root$fastcgi_script_name;
32         include fastcgi_params;
33     }
34
35     # Deny access to hidden files
36     location ~ /\. {
37         deny all;
38     }
39
40     # Deny access to config files
41     location ~ /config {
42         deny all;
43     }
44
45     # Allow public storage access
46     location /storage {
47         alias /var/www/storage/app/public;
48         expires 1y;
49         add_header Cache-Control "public, immutable";
50     }

```

Bukti hasil:



Link Repository: [Repository Laravel Day 14](#)